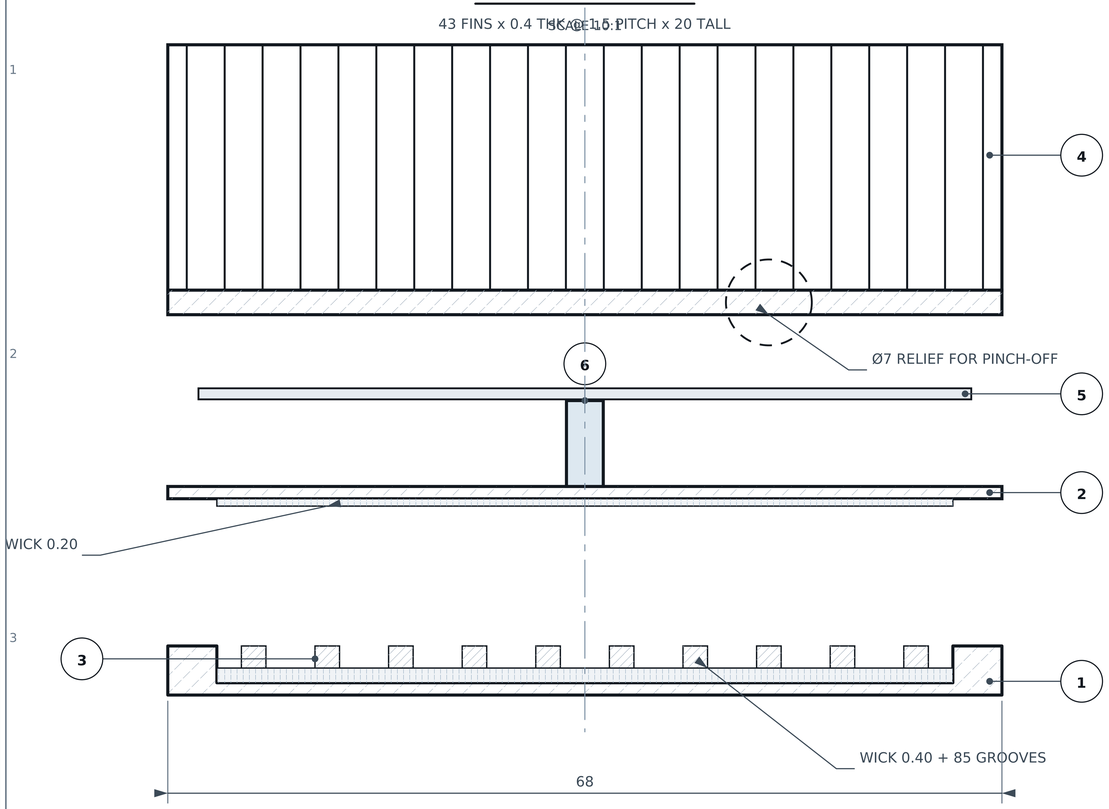
EXPLODED ASSEMBLY

SCALE 10:1



WICK THICKNESS SHOWN x3 FOR CLARITY / ASSEMBLED HEIGHT 27.0

THERMAL RESISTANCE NETWORK

JUNCTION TO AIR, 20 x 20 DIE

DIE	TIM	Cu FLOOR	EVAP	VAPOUR	COND	LID+SPRD	TIM	FIN→AIR	AIR
—	0.025	0.006	0.100	0.004	0.012	0.036	0.011	0.470	25 C

SERIES TOTAL R th(j-a) = 0.67 K/W @ 2 m/s → Q = 113 W at Tj 100 C / air 25 C

Vapour chamber contributes 0.194 K/W (29%); the fin stack dominates. Evaporator wick is the largest single internal term — do not thin the floor wick below 0.40.

BILL OF MATERIALS — ASSEMBLY VC-100

ITEM	QTY	PART No.	DESCRIPTION	MATERIAL	MASS
1	1	VC-101	CHAMBER BASE, POCKETED, 100 POSTS	Cu C1100 OFHC	77.4 g
2	1	VC-102	CHAMBER LID, 1.0 THK, FILL PORT	Cu C1100 OFHC	41.4 g
3	1	VC-103	SINTERED WICK — IN-SITU PROCESS	Cu POWDER 45-75um	9.6 g
4	1	VC-201	FIN STACK, 43 FIN — FORCED AIR	Al 1100-H14	88.1 g
4A	1	VC-202	FIN STACK, 13 FIN — NAT. CONVECTION	Al 1100-H14	72.7 g
5	1	VC-204	THERMAL INTERFACE PAD 0.2 THK	PCM / SIL-PAD	0.4 g
6	1	VC-205	PINCH-OFF TUBE OD3.0 x ID2.0 x 25	Cu C1100	1.2 g
7	1.90 mL	VC-206	WORKING FLUID — DEAERATED	DI H2O 18 MOhm.cm	1.87 g

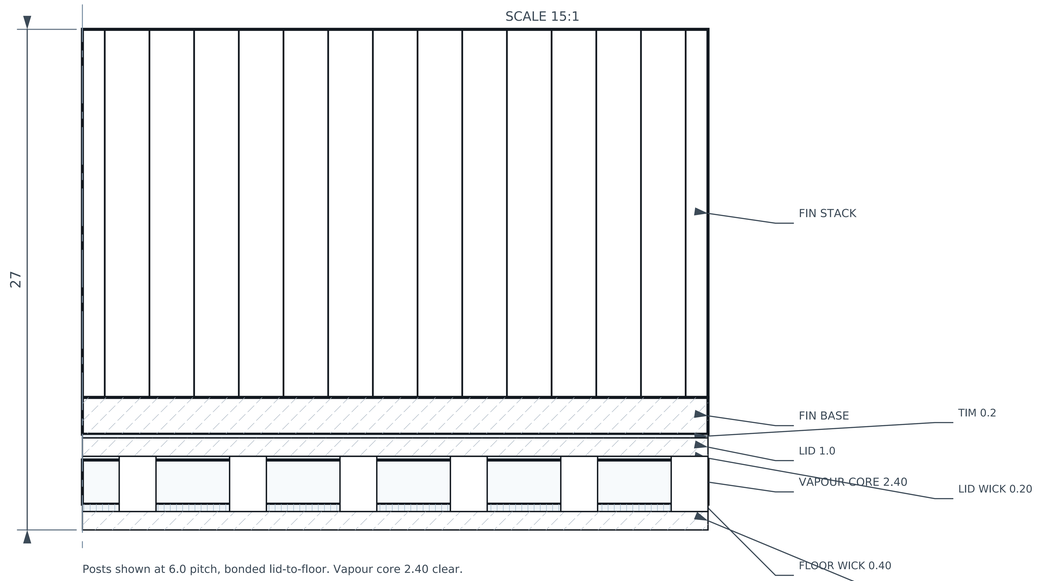
TOTAL MASS, CHARGED: 218 g (CONFIG A) / 203 g (CONFIG B)

PERFORMANCE — 20 x 20 DIE, Tj 100 C, AIR IN 25 C

CONFIGURATION	AIR	R th (j-a)	Q @ dT 75 K
CONFIG A — 43 fin @ 1.5 pitch	1 m/s	1.12 K/W	**67 W**
CONFIG A — 43 fin @ 1.5 pitch	2 m/s	0.67 K/W	**113 W**
CONFIG A — 43 fin @ 1.5 pitch	4 m/s	0.46 K/W	**163 W**
CONFIG B — 13 fin @ 5.0 pitch	natural	3.42 K/W	22 W
CONFIG B — passive, Tj 85 C limit	**natural**	**—**	**16.5 W**
capillary limit — water, vertical	—	—	804 W
capillary limit — water, horizontal	—	—	858 W

SECTION X—X (HALF, FROM CENTRELINE)

SCALE 15:1



Posts shown at 6.0 pitch, bonded lid-to-floor. Vapour core 2.40 clear.

GENERAL NOTES

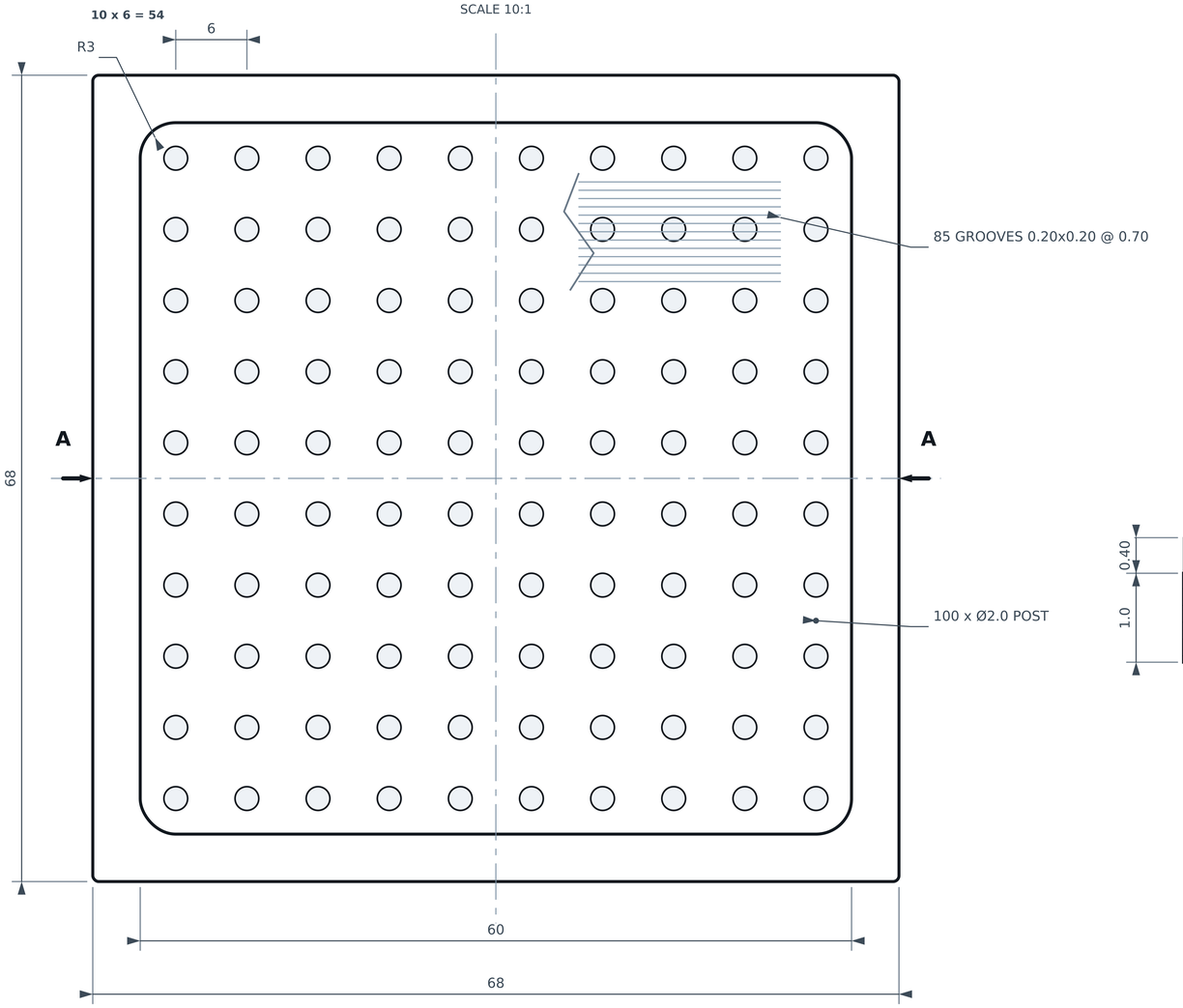
1. Interpret per ISO 128. Tolerances ISO 2768-mK unless stated. Dimensions in mm.
2. Charging, evacuation and sealing per VC-300. NEVER charge before diffusion bonding — bond temperature exceeds the fluid critical pressure envelope.
3. CONFIG A requires forced air >= 1 m/s. For still air use CONFIG B (VC-202); a 1.5 mm fin pitch chokes natural convection to h = 0.03 W/m2K.
4. Orientation-independent: 10.5 kPa capillary head vs 0.66 kPa gravity head over 68 mm.

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **GENERAL ASSEMBLY** — VAPOUR CHAMBER + FIN STACK / Cu-H2O / 68 x 68 x 27	SCALE: **10:1**	UNITS: **mm**	THIRD ANGLE PROJECTION
DRAWING No.: **VC-100**	REV: **A**	SHEET: **1 / 7**	
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES	DATE: 2026-09-09

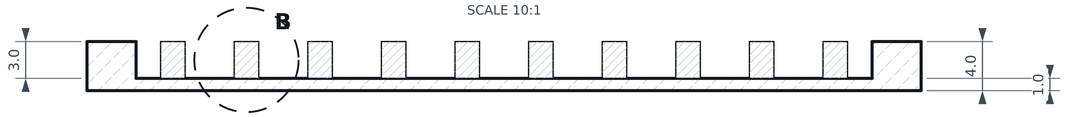
PLAN — POCKET SIDE UP

SCALE 10:1



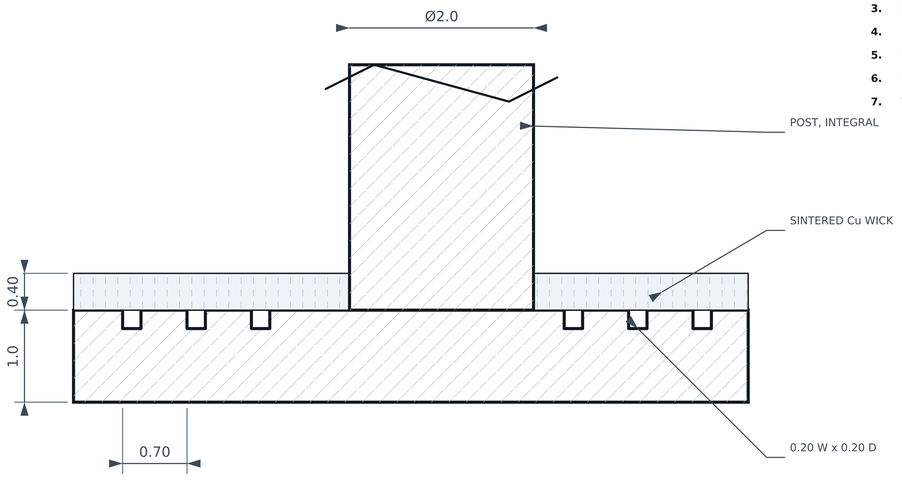
SECTION A—A

SCALE 10:1



DETAIL B

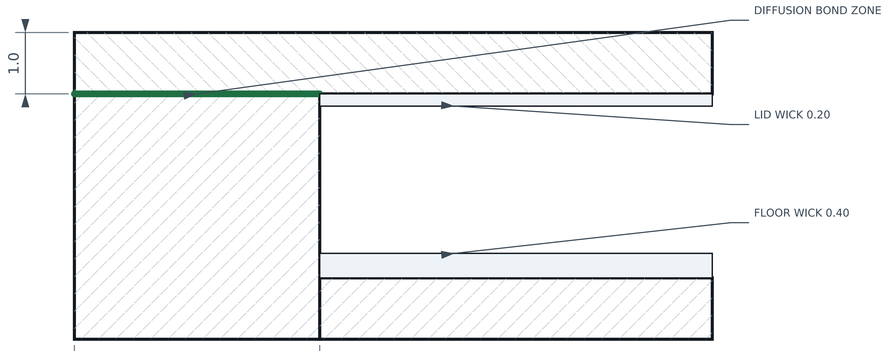
SCALE 75:1



DETAIL C — RIM BOND LAND

SCALE 50:1

Bond area 4.0 x 4 sides = 1024 mm². Burst margin > 16 at 150 C.



NOTES

1. MATL: Cu C1100 / C10100 OFHC, annealed.
2. Posts integral with base — machine from solid or coin.
3. Bond faces (rim + post tops): flatness 0.02, Ra 0.4 max.
4. Grooves run ONE direction only, interrupted at posts.
5. Post top faces coplanar with rim within 0.015.
6. Deburr 0.1 max. No burrs into cavity.
7. Vacuum degrease + acid pickle before sintering.

CRITICAL CHARACTERISTICS

FEATURE	NOMINAL	TOL	CRITICAL TO
Floor thickness	1.0	+0.05 / -0.00	evaporator R th
Post height = pocket depth	3.0	±0.015	lid bond / burst
Rim + post-top flatness	—	0.02	hermeticity
Groove depth	0.20	±0.03	capillary return
Bond face roughness	Ra 0.4	max	diffusion bond
Post position	true position	Ø0.10 MMC	lid land alignment

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **CHAMBER BASE**

Cu C1100 (OFHC) / 68 x 68 x 4.0 / 100 INTEGRAL POSTS

SCALE: **10:1** — THIRD ANGLE PROJECTION

UNITS: **mm**

DRAWING No.: **VC-101**

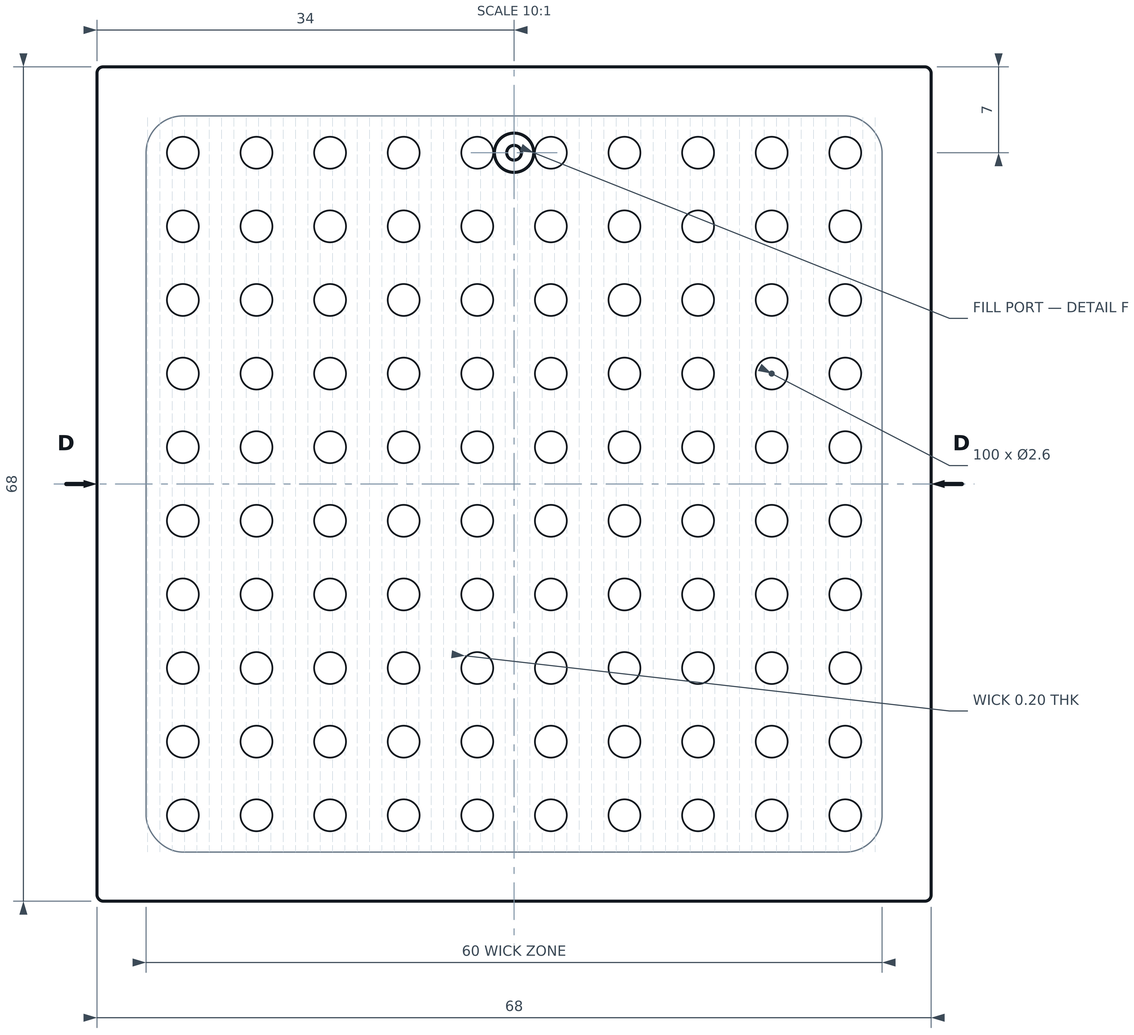
REV: **A**

SHEET: **2 / 7**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-09

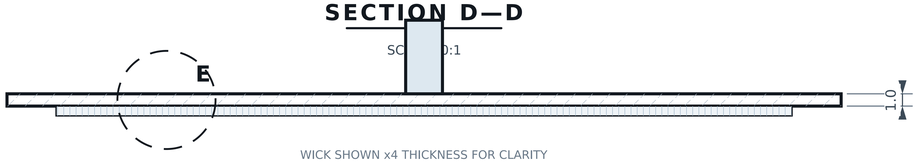
PLAN — WICK SIDE UP

SCALE 10:1



SECTION D—D

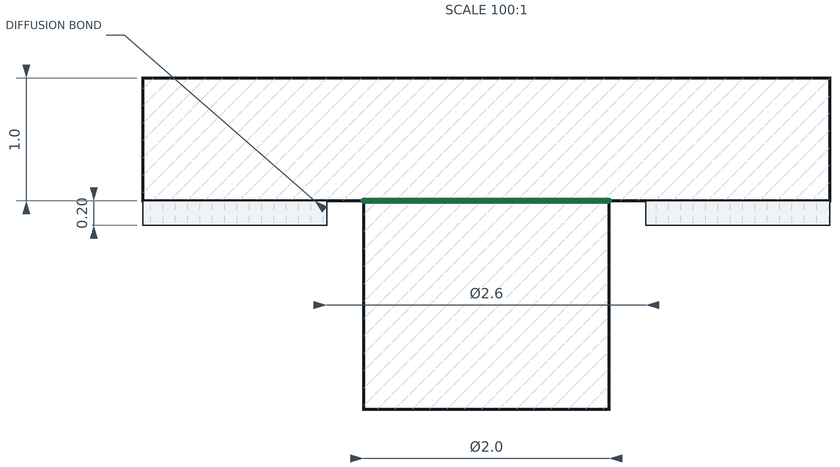
SCALE 10:1



WICK SHOWN ×4 THICKNESS FOR CLARITY

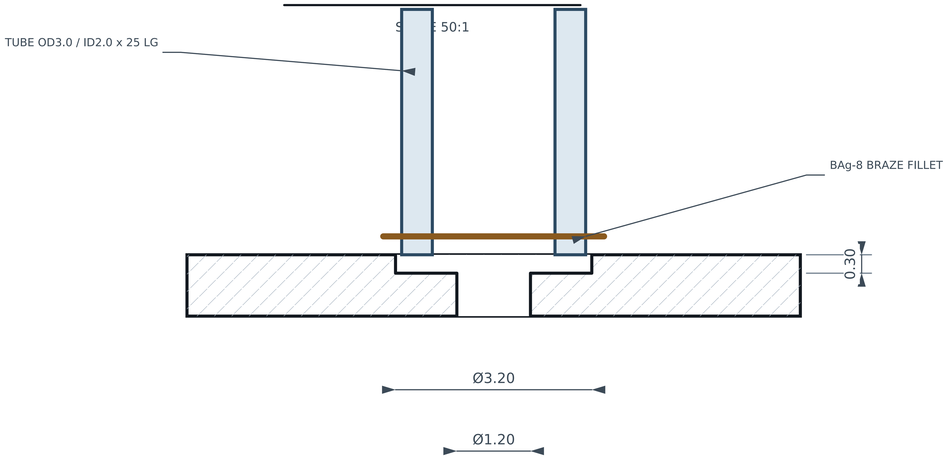
DETAIL E — POST LAND

SCALE 100:1



DETAIL F — FILL PORT

SCALE 50:1



Port sits inside the pocket footprint, clear of posts by 2.0 min.

Pinch off, cold-weld, then laser seal the tip. Do NOT use a screw seal.

NOTES

1. MATL: Cu C1100 OFHC annealed, 1.0 sheet.
2. Flatness 0.02 over 68 x 68 after sintering.
3. Wick clearance holes keep post lands bare for metal-to-metal bond.
4. Braze tube before bake-out, after diffusion bonding.
5. Handle with gloves. No hydrocarbons on bond faces.

INSPECTION

CHECK	REQUIREMENT	METHOD
Flatness after sinter	0.02 max	CMM / optical flat
Wick adhesion	no spall after 3 thermal cycles	visual 10x
Post land bare area	Ø2.6 min clear	optical
Braze joint	He leak < 1e-9 Pa.m3/s	mass spectrometer

FILL P

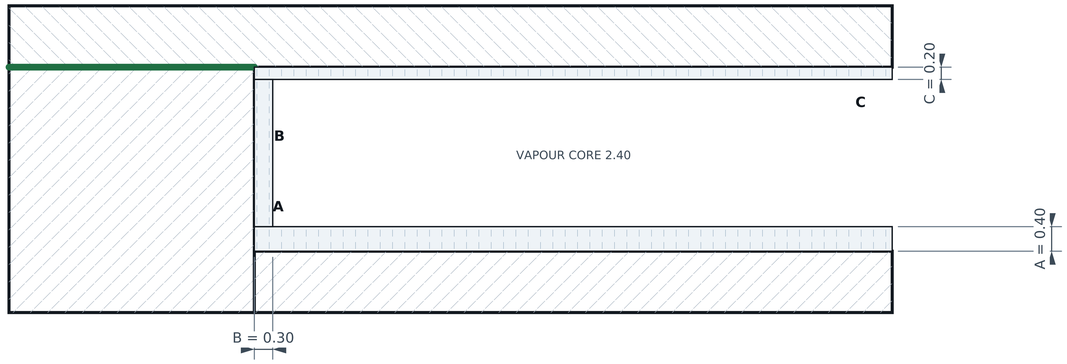
l port p
earanc
Counterb
ube pro

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **CHAMBER LID** — Cu C1100 (OFHC) / 68 x 68 x 1.0 / WICK-LINED, FILL PORT	SCALE: 10:1	UNITS: mm	THIRD ANGLE PROJECTION
DRAWING No.: VC-102	REV: A	SHEET: 3 / 7	
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES
DATE: 2026-09-09			

WICK ZONE SECTION

SCALE 50:1 — ASSEMBLED



Zones A, B and C are sintered in ONE cycle and must be capillary-continuous:

condensate returns C → B → A. A break at the B/A junction causes evaporator dry-out.

WICK ZONES

ZONE	LOCATION	THK	FUNCTION
A	Pocket floor, around posts	0.40	evaporator + liquid supply
B	Pocket side walls (4 off)	0.30	perimeter return path
C	Lid underside, posts masked	0.20	condensate collection

SINTER PROCESS

PARAMETER	VALUE	TOL / NOTE
Cu powder, spherical, gas-atomised	**45 – 75 um**	sieved, -200/+325 mesh
Effective pore radius	**12.6 um**	= 0.21 x d(particle)
Porosity	**55 %**	± 5 %, quantitative metallography
Permeability (Blake-Kozeny)	1.97e-11 m2	derived, not measured
Capillary head, water @ 60 C	**10.5 kPa**	vs 0.66 kPa gravity head
Sintering temperature	950 C	± 10 C
Dwell	30 min	at temperature
Atmosphere	H2 (dew pt < -40 C)	or vacuum < 1e-3 Pa
Ramp / cool	10 C/min / furnace cool	no quench
Tooling	graphite mandrel + mask	masks post tops, rim, fill port

WICK ACCEPTANCE

ACCEPTANCE TEST	REQUIREMENT	METHOD
Porosity	55 % ± 5	image analysis, 3 sections
Wick thickness A	0.40 ± 0.05	cross-section, 5 points
Capillary rise, water	> 25 mm in 60 s	witness coupon, vertical
Groove continuity	open, no blockage	back-light / CT, first article

POWDER TRADE STUDY — WHY 45-75 um

POWDER	r eff	dP cap	Q cap VERT	VERDICT
25 um	5.2 um	25.2 kPa	348 W	clogs, low K
45 – 75 um	**12.6 um**	**10.5 kPa**	**804 W**	**SELECTED**
100 um	21.0 um	6.3 kPa	1279 W	coarse, ok
150 um	31.5 um	4.2 kPa	1802 W	marginal tilt
400 um (original drawing)	100 um	0.4 kPa	0 W	FAILS — < gravity

The original 200 um pore spec gives 380 Pa of capillary head against a 500 Pa gravity head

over 68 mm: the wick cannot lift liquid at all in any vertical orientation.

CAPILLARY PRESSURE BUDGET — WORST CASE, VERTICAL

TERM	VALUE	NOTE
Capillary head available 2σ/r	**10 508 Pa**	water @ 60 C, r = 12.6 um
Liquid loss through wick + grooves	− 12.6 Pa/W	at 804 W: 10 127 Pa
Vapour loss through 2.40 core	− 0.2 Pa/W	negligible
Gravity head, 68 mm adverse	− 656 Pa	vertical, evaporator up
Margin at 113 W design point	**+ 8 428 Pa**	**7.4 x**

The wick is sized by the vertical case. Horizontal operation carries 858 W and is never the limit.

PROCESS NOTES

1. Mask post top faces and the 4.0 rim before sintering — sintered powder on a bond land destroys hermeticity.
2. Mask the fill port bore. Powder in the pinch-off tube prevents a gas-tight cold weld.
3. Grooves must remain open under the wick. Verify by back-light or CT on the first article.
4. Sinter and diffusion bond may share one furnace cycle only if the bond fixture applies load after the sinter dwell; otherwise run separately.
5. No organic binders. Any residue outgasses into the sealed chamber as non-condensable gas.

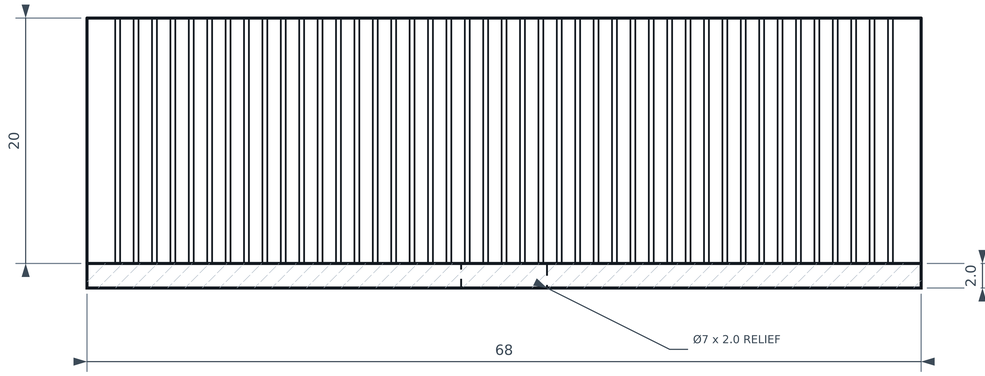
INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **SINTERED WICK — PROCESS SPECIFICATION** (IN-SITU SINTERED Cu POWDER / ALL WICK ZONES)	SCALE: **50:1** (THIRD ANGLE PROJECTION)	UNITS: **mm**	
DRAWING No.: **VC-103**	REV: **A**	SHEET: **4 / 7**	
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES / DATE: 2026-09-09

CONFIG A — VC-201

FORCED AIR >= 1 m/s

43 FINS x 0.40 THK @ 1.5 PITCH

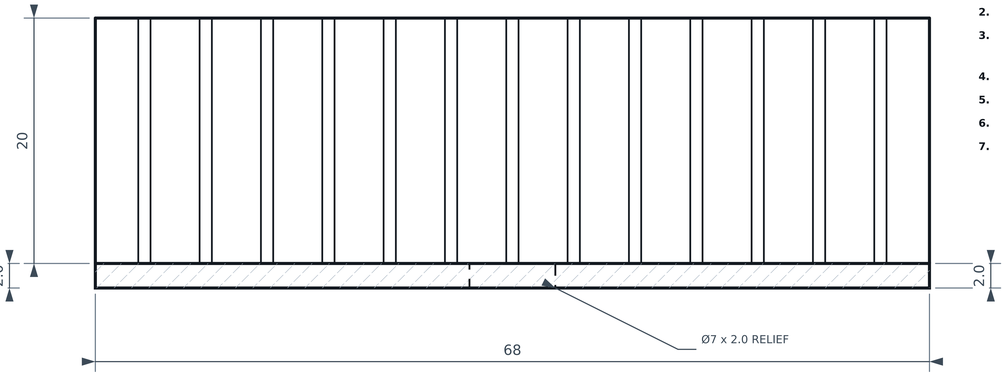


FRONT ELEVATION

CONFIG B — VC-202

NATURAL CONVECTION ONLY

13 FINS x 1.00 THK @ 5.0 PITCH



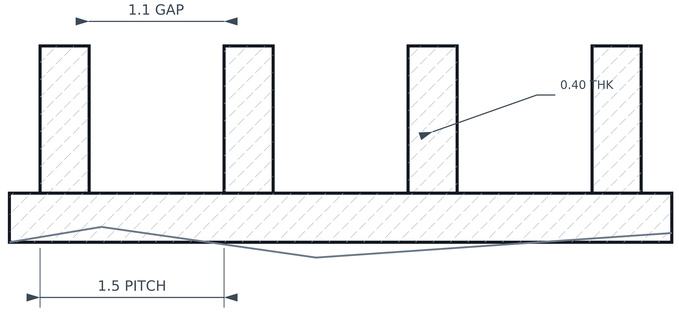
FRONT ELEVATION

NOTES

1. MATL: Al 1100-H14 or 6063-T5. Skived, folded or bonded fin.
2. Base flatness 0.03. Ra 0.8 max on the TIM face.
3. VC-202 SHALL be anodised, 5 um clear minimum. Radiation carries 28% of the passive load.
4. VC-201 finish is free - forced convection swamps radiation.
5. Fit ONE configuration only - alternates, not stackable.
6. Relief pocket clears the folded pinch-off tube, 0.5 min after pinch.
7. Fin tips deburred. No swarf.

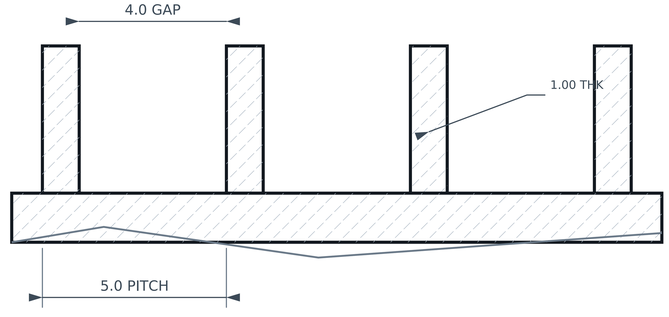
FIN DETAIL

SCALE 100:1



FIN DETAIL

SCALE 30:1



PASSIVE CAPACITY - VC-202, Tj 85 C MAX, NO FAN, FINS VERTICAL

AMBIENT	dT	FIN BASE	CONVECTION	RADIATION	TOTAL
25 C	60 K	82 C	11.9 W	4.6 W	**16.5 W**
30 C	55 K	82 C	10.5 W	4.3 W	14.8 W
35 C	50 K	82 C	9.1 W	4.0 W	13.1 W
40 C	45 K	83 C	7.7 W	3.7 W	11.4 W
45 C	40 K	83 C	6.4 W	3.4 W	9.8 W

WHY THE FINISH IS SPECIFIED - Tj 85 C, 25 C AMBIENT

FIN FINISH	eps	RAD	TOTAL	GAIN
bare mill finish	0.05	0.3 W	12.4 W	1.00x
chromate conversion	0.60	3.3 W	15.2 W	1.23x
clear anodise 5 um (SPECIFIED)	0.85	4.6 W	**16.5 W**	1.33x
black anodise	0.92	5.0 W	16.9 W	1.36x

Pitch optimum 4.8 mm (17.0 W); 5.0 mm used as a round tooling pitch.

Passive floor with channels fully blocked: 4.6 W from envelope radiation alone.

SELECTION TABLE - Tj 85 C ROWS AT 25 C AMBIENT

	PITCH	FINS	FINISH	Q @ Tj 100 C	Q @ Tj 85 C
VC-201 - forced air 1 m/s	**1.5**	**43**	any	**67 W**	**54 W**
VC-201 - forced air 2 m/s	**1.5**	**43**	any	**113 W**	**90 W**
VC-201 - forced air 4 m/s	**1.5**	**43**	any	**163 W**	**130 W**
VC-201 - STILL AIR (invalid)	1.5	43	anodised	9 W	7 W
VC-202 - passive, fins VERTICAL	**5.0**	**13**	**anodised**	**22 W**	**16.5 W**
VC-202 - passive, fins horizontal	5.0	13	anodised	13 W	9.5 W
VC-202 - passive, bare aluminium	5.0	13	bare	17 W	12.4 W

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE

FIN STACK — TWO CONFIGURATIONS

Al 1100-H14 / SKIVED OR BONDED / 68 x 68 x 22

SCALE **10:1** — UNITS **mm** — THIRD ANGLE PROJECTION

DRAWING No. **VC-201 / VC-202** — REV **A** — SHEET **5 / 7**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-09

MANDATORY PROCESS SEQUENCE

EVERY STEP IN ORDER — NO SUBSTITUTION

1 — MACHINE

VC-101 base + VC-102 lid. Posts integral. Fill port drilled + counterbored.

2 — CLEAN — no hydrocarbons

Vapour degrease → 10% H2SO4 pickle 60 s → DI rinse → N2 blow dry.

3 — SINTER WICK — per VC-103

Cu powder 45-75 um. 950 C / 30 min / H2 or vacuum. Mask rim, post tops, fill bore.

4 — DIFFUSION BOND — Cu-Cu solid state

Stack lid on base. 850 C / 8 MPa / 60 min / vacuum < 1e-3 Pa.

5 — WELD FILL TUBE — after bonding

Laser or micro-TIG weld Cu tube OD3.0 to lid. No filler, no flux.

6 — LEAK TEST 1 — gross + fine

He mass spec, chamber under vacuum. Reject > 1e-9 Pa.m3/s.

7 — BAKE-OUT — removes adsorbed gas

200 C / 4 h / < 1e-3 Pa through the fill tube. Pump throughout.

8 — CHARGE — ± 0.05 mL

1.90 mL DI water 18 MOhm.cm, deaerated. Burette under vacuum, chamber at 0 C.

9 — DEGAS — removes dissolved O2

Freeze-pump-thaw x 3. Final pump at ice point to < 10 Pa.

10 — PINCH-OFF — NO screw seal

Cold-weld the tube in a hydraulic pincher. Laser-seal the tip. Trim + fold.

11 — LEAK TEST 2 — post-seal

He bomb 3 bar / 4 h, then mass spec. Reject > 1e-9 Pa.m3/s.

12 — BURN-IN — stabilises wick

3 thermal cycles 20 ↔ 100 C. Re-test R th after.

13 — ACCEPT — final gate

R th (case-to-case) < 0.25 K/W at 50 W, horizontal and vertical.

PROCESS PARAMETERS

PARAMETER	VALUE	LIMIT
Working fluid	**DI water 18 MOhm.cm**	deaerated
Charge volume	**1.90 mL**	± 0.05 mL
Wick void volume	1.51 mL	charge = 125 %
Bake-out	200 C / 4 h	< 1e-3 Pa
Final vacuum before seal	< 10 Pa	at ice point
Diffusion bond	850 C / 8 MPa / 60 min	vacuum
Leak rate	**< 1e-9 Pa.m3/s**	He mass spec
Acceptance R th	**< 0.25 K/W @ 50 W**	both orientations

PROHIBITED — CAUSES FIELD FAILURE

- Charging before diffusion bonding. Water boils at 100 C and the bond runs at 850 C: the chamber ruptures. This error appears on the superseded Maxsorb drawing — do not repeat it.
- Threaded or O-ring fill seals. A screw seal cannot hold high vacuum; helium ingress accumulates as non-condensable gas and the condenser is blanketed within months. Cold-weld pinch-off only.
- Aluminium anywhere wetted. Al + water generates H2. The fin stack is external and TIM-coupled only.
- Undeaerated water. Dissolved O2 oxidises copper and liberates H2.
- Activated carbon or any organic inside the envelope without a 250 C / 8 h vacuum bake.

Non-condensable gas is the dominant failure mode of a sealed two-phase device.

Steps 7, 9 and 10 exist solely to prevent it. None of them is optional.

QUALITY RECORDS

STEP	RECORD RETAINED	HOLD POINT
4	bond furnace chart (T, P, vacuum)	yes — first article
6	He leak rate, pre-charge	**yes — 100 %**
7	bake-out pressure trace	yes
8	charge mass, by weight	**yes — 100 %**
11	He leak rate, post-seal	**yes — 100 %**
13	R th, both orientations	**yes — 100 %**

Charge by MASS, not by volume: 1.90 mL DI water = 1.87 g at 20 C. A burette

reading is not traceable once the chamber is cold and under vacuum.

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **PROCESS & CHARGING SPECIFICATION**
Cu-H2O VAPOUR CHAMBER / MANDATORY SEQUENCE

SCALE: NONE — THIRD ANGLE PROJECTION
UNITS: mm

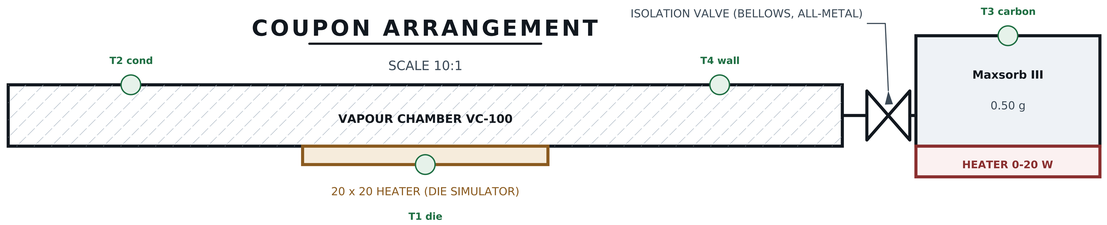
DRAWING No.: VC-300
REV: A
SHEET: 6 / 7

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-09

EXPERIMENTAL COUPON — NOT A PRODUCTION CONFIGURATION — SEE PHYSICS NOTE BELOW

COUPON ARRANGEMENT

SCALE 10:1



Mesh: 10 um SUS 316L — NOT 50 um. Maxsorb III fines pass a 50 um mesh and foul the wick.

Carbon bake-out 250 C / 8 h / < 1e-3 Pa before installation, transferred under vacuum.

Reservoir 0.5 g Maxsorb III needs >= 1.8 cm3 at 0.28 g/cc bulk density — the original 1.50 cm3

housing could not physically hold the charge. This coupon uses 30 x 12 x 6 = 2.16 cm3.

ADSORPTION EQUILIBRIUM — DUBININ-ASTAKHOV, METHANOL / MAXSORB III

T evap	T res	P/Psat	UPTAKE	HELD	WICK FILL	RESULT
60 C	40 C	1.00	1.31 g/g	0.66 g	15 %	**DRY**
60 C	60 C	1.00	1.31 g/g	0.66 g	15 %	**DRY**
60 C	100 C	0.24	1.06 g/g	0.53 g	35 %	**DRY**
60 C	127 C	0.11	0.77 g/g	0.39 g	58 %	marginal
60 C	169 C	0.03	0.42 g/g	0.21 g	90 %	wet

PHYSICS NOTE — WHY THE PASSIVE VERSION CANNOT WORK

- In a sealed pure-substance chamber the pressure is set by the evaporator: P = Psat(T_evap). The carbon sees the relative pressure P / Psat(T_res).
- A reservoir on the condenser side always satisfies T_res <= T_evap, so P/Psat >= 1 and the carbon saturates: it takes 0.66 g of a 0.75 g charge and the wick runs at 15 % fill. That is permanent dry-out, not pressure control.
- Keeping the wick wet needs the reservoir ~110 K ABOVE the chip. A working vapour chamber is isothermal, so no passive layout can deliver that. It requires the heater and valve drawn above.
- Overcharging to 1.90 mL keeps the wick wet, but then liquid always remains, the pressure stays at Psat, and the carbon buys nothing thermally. Its only genuine residual function is as a non-condensable gas getter — for which tens of mg suffices, not 0.5 g.

COUPON vs PRODUCTION VC-100

	WITH CARBON	WITHOUT
Wick fill at equilibrium	15 %	**125 %**
Q at 2 m/s forced air	0 W (dry)	**113 W**
Transient buffer	722 J (heater needed)	0 J
Parts count / failure modes	+ valve, heater, control	**baseline**

INSTRUMENTATION

MEASUREMENT	INSTRUMENT	PURPOSE
T1..T4	type-T TC, ± 0.2 K	gradient map
Chamber pressure	capacitance manometer	verify Psat(T)
Reservoir mass	in-situ load cell ± 1 mg	uptake vs time
Heater power	4-wire, ± 0.5 %	energy balance
Valve state	open / closed	isolate the carbon

TEST MATRIX

#	TEST	MEASURES
1	Valve CLOSED, sweep 0-150 W	baseline R th, no carbon
2	Valve OPEN, reservoir unheated	dry-out rate + time constant
3	Valve OPEN, reservoir held 110 K above die	whether the wick recovers
4	Valve OPEN, cyclic 10 W bursts, heater regen	usable buffer energy (722 J predicted)
5	1000 h hold, valve OPEN, unheated	NCG getter benefit vs dry-out cost

COUPON NOTES

- Fluid is METHANOL for this coupon (activated carbon is hydrophobic; water uptake is poor). Accept the 7.3x lower merit number — this coupon measures adsorption, not peak cooling.
- Methanol charge 1.90 mL so the wick stays wet with the carbon saturated. Do not run at 1.00 mL.
- The valve is what makes this an experiment rather than a defect. A production unit has no valve, so it has no way to stop the carbon draining the wick.

IF TEST 4 SUCCEEDS — PATH TO A PRODUCT

- Move the reservoir onto the die's own heat path, upstream of the chamber, so it is genuinely hotter than the evaporator without a valve.
- Replace the bellows valve with a thermally actuated one-way element, or accept a control loop driving the regeneration heater.
- Budget 722 J against ~5 g of added mass and a new hermetic joint. Compare against simply adding 15 K of copper (+50 g).

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE
MAXSORB III RESERVOIR — TEST COUPON
EXPERIMENTAL / NOT FOR PRODUCTION / INSTRUMENTED

SCALE **10:1** — THIRD ANGLE PROJECTION
UNITS **mm**

DRAWING No. **VC-400**
REV **A**
SHEET **7 / 7**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-09